

Learned Regularization for Robust Image Reconstruction: A Comparative Study of Plug-and-Play Priors under Noise, Undersampling, Model Mismatch, and Distribution Shift

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Abstract

We study sparse-view computed tomography as a linear inverse problem and compare classical variational regularization against a learned image prior integrated into an optimization-based reconstruction framework. Specifically, we train a noise-level-conditioned CNN denoiser and use it as an implicit prior inside a Plug-and-Play proximal-gradient (PnP-PGD) scheme, benchmarking it against filtered back-projection, Tikhonov (L2), and total variation (TV) regularization. Using a controlled evaluation protocol – disjoint train/validation/test splits, per-condition hyperparameter tuning, and paired statistical testing – we find that the learned prior improves PSNR by up to 4 dB over TV under high measurement noise or severe angular undersampling, but underperforms TV in the near noise-free regime. A convergence study reveals that PnP-PGD does not converge to a fixed point that is simultaneously data-consistent and maximally accurate: reconstruction quality peaks after a handful of iterations and then degrades, while the data-fidelity residual continues to rise. We further show that the learned prior’s advantage narrows under forward-model mismatch (misspecified projection angles) and reverses under train-test distribution shift, where the prior hallucinates in-distribution structure on out-of-distribution inputs. We discuss these findings in light of known instability results for learned reconstruction methods and argue that robustness evaluation should be a standard, not optional, component of learned-prior reconstruction studies.

1 Introduction

Image reconstruction from indirect, noisy, and incomplete measurements is a classical ill-posed inverse problem, arising in computed tomography (CT), magnetic resonance imaging, and many other imaging modalities. Given a forward operator A and noisy measurements

$$y = Ax + n, \tag{1}$$

the goal is to recover x from y when A is ill-conditioned or rank-deficient, as is the case for sparse-view CT with few projection angles. Classical approaches address ill-posedness through explicit regularization, solving

$$\hat{x} = \arg \min_x \frac{1}{2} \|Ax - y\|_2^2 + \lambda R(x), \tag{2}$$

with R a hand-crafted prior such as Tikhonov (L2) or total variation (TV) penalties. In recent years, learned image priors – most notably through Plug-and-Play (PnP) methods and deep unrolling

– have been proposed to replace or augment hand-crafted regularizers with a denoising neural network, exploiting the empirical observation that image denoisers implicitly encode rich natural-image statistics [6].

While learned priors frequently outperform classical regularization on in-distribution benchmarks, a growing body of work has raised concerns about their *robustness*: sensitivity to distribution shift, lack of convergence guarantees, and instabilities under small perturbations of the input or forward model [1]. Many project-level or introductory studies of learned reconstruction report only in-distribution accuracy, leaving these robustness questions unaddressed.

This paper documents a controlled, small-scale study designed explicitly to probe these questions. We deliberately restrict scope to a simple, cheaply reproducible setting – sparse-view tomography of 28×28 MNIST-derived images – in order to run a thorough grid of experiments with proper train/validation/test protocol rather than a single headline comparison. Our contributions are:

- A from-scratch implementation of a sparse-view Radon operator with exact adjoint, classical regularization baselines (FBP, Tikhonov, TV-ADMM), and a PnP-PGD solver using a noise-conditioned DnCNN prior.
- A systematic comparison of PSNR/SSIM accuracy across measurement noise levels and numbers of projection angles, with validation-tuned hyperparameters and paired significance testing.
- A convergence analysis explaining why PnP-PGD selects very few iterations under validation tuning, showing a peak-then-decay pattern in reconstruction accuracy that is not visible from residual curves alone.
- Two robustness studies – forward-model mismatch (angle offset and jitter) and train-test distribution shift (Fashion-MNIST, EMNIST, and a CT phantom) – run with hyperparameters frozen at their nominal values, exposing failure modes not visible under in-distribution evaluation alone.

2 Related Work

Variational regularization. Tikhonov regularization and total variation [4] remain standard baselines for ill-posed inverse problems due to their convexity, well-understood convergence properties, and interpretability. TV in particular is a strong prior for piecewise-constant images, which makes it a competitive baseline for digit-like data such as MNIST.

Plug-and-Play priors. Venkatakrishnan et al. [6] introduced the idea of replacing the proximal operator of a variational regularizer with an off-the-shelf denoiser inside an ADMM or proximal-gradient iteration. Subsequent work analyzed convergence conditions for PnP under assumptions on the denoiser (e.g. non-expansiveness or bounded Lipschitz constant) [2, 5]; without such constraints, plain CNN denoisers offer no convergence guarantee, a point directly relevant to the convergence behavior we observe in Section 4.

Learned denoisers. DnCNN [7] demonstrated that residual CNNs trained for Gaussian denoising across a range of noise levels generalize well as plug-in priors. We adopt this noise-conditioning strategy so that a single trained network can serve across the PnP iteration trajectory, where the effective denoising strength is a tunable hyperparameter.

Instabilities and robustness. Antun et al. [1] showed that learned reconstruction methods for CT and MRI can be unstable to small perturbations and structural changes not present in training data, in some cases more so than classical methods. Related work on distribution shift in learned inverse problem solvers [3] similarly finds that accuracy gains from learned priors can shrink or reverse under shift. Our distribution shift and forward-model mismatch experiments are a small-scale, project-level instantiation of this broader concern, and our findings (Section 5) are consistent with it.

3 Methodology

3.1 Forward operator

We construct a sparse-view Radon transform $A \in \mathbb{R}^{m \times n}$ as an explicit dense matrix by applying the discrete Radon transform to each canonical basis image and stacking the resulting sinogram columns. This gives an exact adjoint $A^\top$ (rather than an approximate one, as is common with implicit/fast Radon implementations), which is important for the PnP gradient step, and allows the reconstruction operator to be perturbed independently of the data-generating operator for the mismatch experiments in Section 4. Images are 28×28 ; the number of projection angles and, where relevant, an angular offset or per-angle jitter are configurable.

3.2 Classical baselines

Filtered back-projection (FBP) applies the standard ramp filter and inverse Radon transform. **Tikhonov regularization** solves

$$\hat{x} = \arg \min_x \frac{1}{2} \|Ax - y\|_2^2 + \frac{\lambda}{2} \|x\|_2^2 = (A^\top A + \lambda I)^{-1} A^\top y, \quad (3)$$

computed in closed form. **Total variation** solves

$$\hat{x} = \arg \min_x \frac{1}{2} \|Ax - y\|_2^2 + \lambda \text{TV}(x) \quad (4)$$

via ADMM with an isotropic TV penalty and a finite-difference operator D , splitting $z = Dx$ and applying the standard shrinkage update on the split variable.

3.3 Learned prior: PnP proximal gradient

We train a DnCNN denoiser D_σ conditioned on the noise level σ (following the FFDNet-style noise-map input), on clean MNIST digits corrupted with Gaussian noise $\sigma \sim \mathcal{U}(0, \sigma_{\max})$, $\sigma_{\max} = 0.3$, using residual learning and an ℓ_2 loss. A single trained network is then used across all PnP experiments, with σ_d (the denoising strength argument passed at inference) treated as a tunable hyperparameter distinct from the training distribution’s noise range.

Reconstruction proceeds by proximal-gradient iteration, replacing the proximal operator of an explicit regularizer with the learned denoiser:

$$x_{k+1} = D_{\sigma_d}(x_k - \eta A^\top (Ax_k - y)), \quad \eta = 1/L(A^\top A), \quad (5)$$

where $L(A^\top A)$ is the largest eigenvalue of $A^\top A$, estimated by power iteration. The iteration is initialized at the FBP reconstruction. Algorithm 1 summarizes the procedure.

Algorithm 1 PnP proximal gradient (PnP-PGD)

Require: measurements y , operator A , denoiser D , strength σ_d , step η , iterations K

```
1:  $x_0 \leftarrow \text{FBP}(y)$ 
2: for  $k = 0, \dots, K - 1$  do
3:    $g_k \leftarrow A^\top(Ax_k - y)$ 
4:    $x_{k+1} \leftarrow D_{\sigma_d}(x_k - \eta g_k)$ 
5: end for
6: return  $x_K$ 
```

3.4 Evaluation protocol

Data are split into disjoint training, validation, and test subsets. All hyperparameters – λ for Tikhonov and TV, and (σ_d, K) for PnP – are selected by grid search maximizing mean validation PSNR, independently for each experimental condition (noise level or number of angles) in Experiments 1–2, and then evaluated once on the disjoint test set. For the robustness experiments (Section 4, E3–E4), hyperparameters are tuned once at a single nominal condition and then *frozen* across all mismatch/shift conditions, reflecting a realistic deployment scenario where a reconstruction pipeline is not retuned for every possible operating condition. We report PSNR and SSIM as mean with 95% confidence intervals over test images, and use a paired Wilcoxon signed-rank test (matched test images and noise realizations) to assess the significance of PnP vs. TV differences.

4 Experiments

4.1 E1–E2: In-distribution accuracy under noise and undersampling

Table 1 reports test-set PSNR for representative conditions (20 projection angles, varying measurement noise σ ; and fixed $\sigma = 0.05$, varying number of angles).

Table 1: Test PSNR (dB) under varying noise and angular undersampling. Best method per row in bold. PnP–TV column reports the paired Wilcoxon test outcome.

Condition	FBP	Tikhonov	TV	PnP-DnCNN	PnP – TV
$\sigma=0$, 20 angles	24.2	28.9	36.0	29.6	-6.4 ($p < 10^{-30}$)
$\sigma=0.01$, 20 angles	23.1	24.1	29.0	29.1	$+0.07$ (n.s., $p = 0.27$)
$\sigma=0.05$, 20 angles	15.5	17.1	20.2	24.0	$+3.85$ ($p < 10^{-30}$)
$\sigma=0.1$, 20 angles	10.4	14.7	16.9	19.9	$+3.0$ ($p < 10^{-30}$)
$\sigma=0.2$, 20 angles	6.6	12.9	14.3	15.3	$+1.0$ ($p < 10^{-10}$)
5 angles, $\sigma=0.05$	9.1	14.2	15.5	17.7	$+2.15$ ($p < 10^{-25}$)
10 angles, $\sigma=0.05$	12.1	15.9	18.4	21.4	$+3.05$ ($p < 10^{-30}$)
90 angles, $\sigma=0.05$	20.6	20.0	24.1	27.1	$+3.04$ ($p < 10^{-30}$)

TV is the strongest classical baseline throughout, consistent with MNIST digits being close to piecewise-constant. The learned prior overtakes TV as soon as the problem departs from the near noise-free, moderately-sampled regime, with gains of up to 4 dB PSNR and a corresponding SSIM improvement (0.66 to 0.94 at $\sigma = 0.05$, 20 angles; full SSIM table in the project repository). At $\sigma = 0$ the learned prior underperforms TV by 6.4 dB, and the two are statistically indistinguishable at $\sigma = 0.01$.

4.2 E5: Why PnP selects few iterations

Validation tuning consistently selects a small number of PnP iterations (5–25 depending on condition). Figure 1 shows PSNR and the data-fidelity residual $\|Ax_k - y\|$ as a function of iteration count, for several values of σ_d , on 50 test images. PSNR rises rapidly and peaks within the first 5–15 iterations, then *decays* to a lower plateau (e.g. $\sigma_d = 0.1$: peak ≈ 24.0 dB near iteration 10, plateau ≈ 22.0 dB by iteration 400). Over the same range, the residual does not decrease; it rises and plateaus, more steeply for larger σ_d . This indicates that the iteration does not converge to a fixed point that is simultaneously data-consistent and maximally accurate: the denoiser’s implicit prior increasingly dominates the data-fidelity term as iterations progress, without a corresponding gain in accuracy. Early stopping is therefore not an artifact of a coarse hyperparameter grid but an active, necessary part of the regularization.

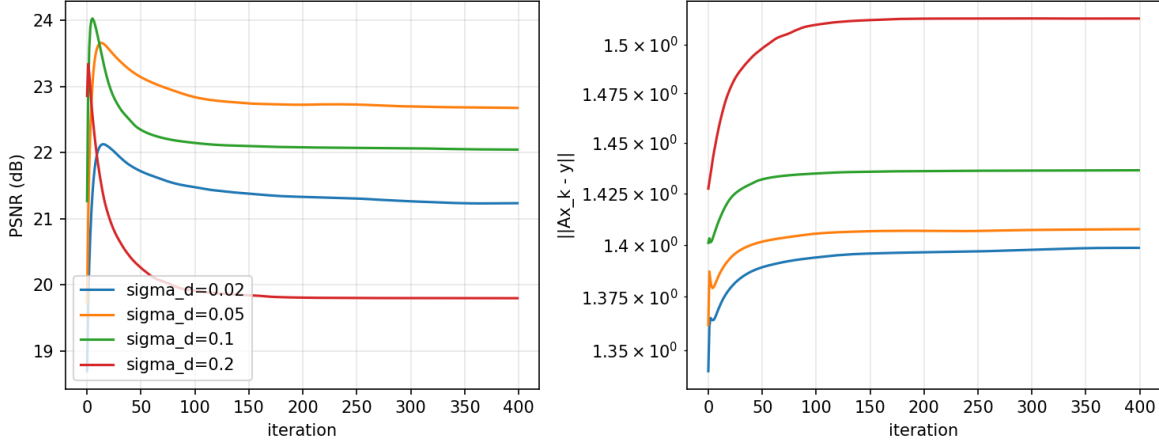


Figure 1: PnP-PGD convergence behavior for several denoising strengths σ_d (50 test images, $\sigma = 0.05$, 20 angles). Left: PSNR vs. iteration, peaking early then decaying. Right: data-fidelity residual vs. iteration, rising and plateauing rather than decreasing.

4.3 E3: Forward-model mismatch

Measurements are generated with the true 20-angle operator; reconstruction uses a perturbed operator with either a constant angle offset (rigid misregistration) or independent per-angle jitter, up to 4° , with all hyperparameters frozen at their $\sigma = 0.05$, 20-angle values. Table 2 summarizes the extremes of the tested range.

Table 2: Test PSNR (dB) under forward-model mismatch, hyperparameters frozen at nominal values.

Mismatch	TV ($0^\circ \rightarrow 4^\circ$)	PnP ($0^\circ \rightarrow 4^\circ$)	TV loss	PnP loss
Constant offset	20.2 $\rightarrow$ 18.6	24.0 $\rightarrow$ 20.0	1.6 dB	4.0 dB
Per-angle jitter	20.2 $\rightarrow$ 19.2	24.0 $\rightarrow$ 22.2	1.0 dB	1.8 dB

PnP remains more accurate than TV in absolute terms throughout the tested range, but degrades substantially faster in relative terms, particularly under a constant angular offset. Its advantage over TV narrows from +3.85 dB at zero mismatch to +1.48 dB at 4° offset.

4.4 E4: Distribution shift

The denoiser is trained exclusively on MNIST digits. We evaluate the full frozen pipeline (nominal $\sigma = 0.05$, 20 angles) on Fashion-MNIST, EMNIST letters, and a resized Shepp-Logan phantom (single image, 20 independent noise draws). Table 3 and Figure 2 summarize the results.

Table 3: Test PSNR (dB) under train–test distribution shift, frozen pipeline.

Test set	TV	PnP-DnCNN	PnP – TV
MNIST (in-distribution)	20.3	24.0	+3.7
EMNIST letters (near shift)	20.1	22.0	+1.9
Fashion-MNIST (far shift)	19.6	16.5	−3.1
Shepp-Logan phantom (far shift)	19.2	15.0	−4.2

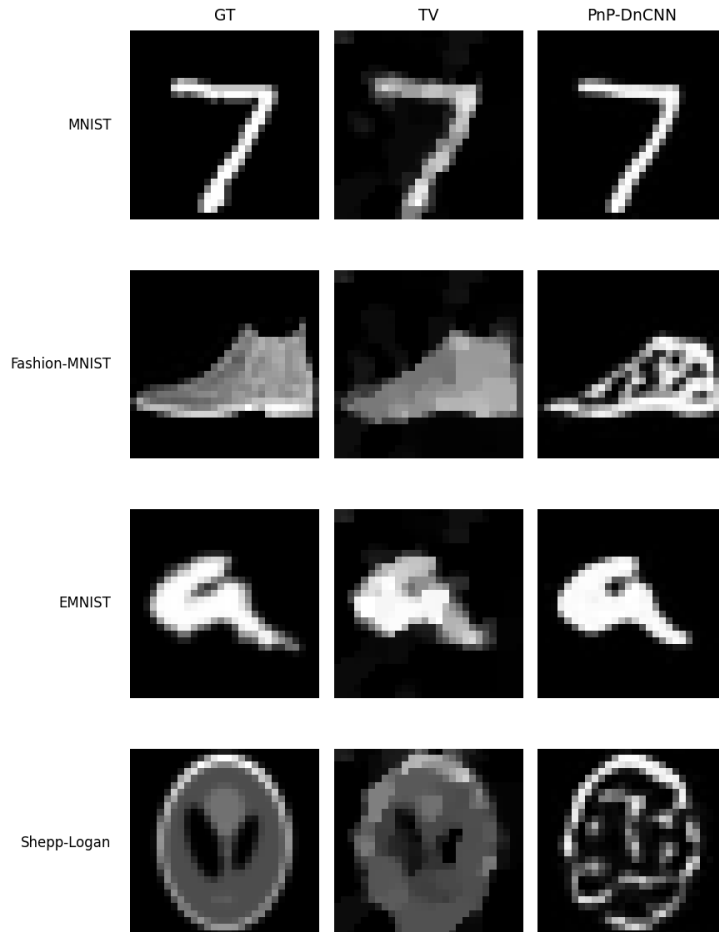


Figure 2: Qualitative reconstructions under distribution shift. The learned prior recovers structure well on MNIST and EMNIST (both stroke-like), but imposes hallucinated digit/stroke-like structure on Fashion-MNIST and the CT phantom, both structurally unlike the training distribution.

The learned prior generalizes to EMNIST letters, which share digits’ stroke-based statistics,

but underperforms TV on both far-shift test sets. Qualitatively (Figure 2), the failure mode is not generic blur but hallucination: the denoiser imposes plausible-looking but incorrect digit/stroke-like structure on inputs unlike its training distribution.

5 Discussion

Four findings emerge from this study. First, the learned prior’s accuracy advantage over TV is conditional, not universal: it is large and significant under noise and undersampling, but reverses in the near noise-free regime, where TV’s piecewise-constant assumption is close to exact for this data. This is a caution against reporting a single aggregate accuracy number for a learned reconstruction method without specifying the operating regime.

Second, the convergence study (Section 4.2) shows that PnP-PGD with an unconstrained CNN denoiser does not behave like a convergent optimization scheme in the classical sense: accuracy peaks early and then degrades even as the algorithm continues to iterate, and the residual does not decrease monotonically. This is consistent with the theoretical observation that PnP convergence guarantees require additional constraints on the denoiser (e.g. non-expansiveness) [5] that plain CNN denoisers, including the one used here, do not satisfy by construction. In this setting, iteration count functions as an implicit, and here empirically necessary, regularization hyperparameter.

Third and fourth, the mismatch and shift experiments show that the learned prior’s advantage is not robust to two distinct kinds of distribution shift: shift in the forward operator, and shift in the image distribution. Under forward-model mismatch the learned method remains more accurate than TV throughout the tested range but loses its advantage faster; under image distribution shift its advantage reverses outright, and does so via a qualitatively concerning failure mode (hallucinated structure) rather than graceful degradation. These results are consistent with broader findings on instabilities in learned reconstruction methods [1] and support the view that in-distribution accuracy alone is an insufficient basis for adopting a learned prior in a reconstruction pipeline; mismatch and shift evaluation should be treated as a standard part of such studies, not an optional addendum.

6 Conclusion

We presented a controlled, small-scale comparison of classical and learned regularization for sparse-view CT reconstruction, evaluated under noise, angular undersampling, forward-model mismatch, and train-test distribution shift. The learned Plug-and-Play prior substantially outperforms total variation regularization in ill-posed, noisy regimes, but this advantage is conditional: it reverses in the near noise-free regime, narrows under forward-model mismatch, and reverses again under distribution shift, where the prior hallucinates in-distribution structure. A convergence analysis further shows that the PnP iteration used here lacks a stable, accurate fixed point, making early stopping an active regularizer rather than an implementation detail. Future work could extend this study to larger, natural-image or clinical-resolution data, provably convergent PnP variants (e.g. with spectrally-normalized or non-expansive denoisers), and real (rather than simulated) forward operators.

Limitations

This study is intentionally small in scale (28×28 MNIST-derived images) to allow a thorough experimental grid; trends may not transfer directly to natural-image or clinical CT resolution and

geometry. The same forward operator is used to generate and reconstruct data in all experiments except E3 (an “inverse crime”), and results reflect a single trained denoiser instance rather than an average over independent training runs. The PnP scheme used has no convergence guarantee, and the forward-model mismatch levels tested (up to 4°) are moderate; larger or structurally different mismatches were not evaluated.

Reproducibility

Code, trained model, all raw result CSVs, and figures are available at: <https://github.com/Abo0raig/lrir.git>.

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